

Total No. of Questions : 10]

SEAT No. :

P2327

[Total No. of Pages : 3

[5254]-662

B.E. (Computer Engineering)
PRINCIPLES OF MODERN COMPILER DESIGN
(2012 Pattern) (Semester - I)

Time : 3 Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Questions. 1 or Q. 2, Q. 3 or Q. 4, Q. 5 or Q. 6, Q. 7 or Q. 8 Q. 9 or Q. 10
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume suitable data, if necessary.
- 4) Figures to the right indicate full marks.

Q1) a) Why compilation phases are divided into front-end and back-end? What are the advantages? [4]

b) Give syntax directed definition for any example arithmetic expression. [6]

OR

Q2) a) What is YACC? Give format of Yacc specification file. [4]

b) Write the syntax directed translation scheme for generating Intermediate code for array assignment statement. [6]

Q3) a) Explain the terms phase and pass related to compiler. [2]

b) Construct LL (1) parsing table for the following grammar. [8]

Terminals = {id, num, while, print, >, {, }, ;, (,)}

Nonterminal = {S, E, B, L}

Rule = i) $S \rightarrow \text{print } (E)$

ii) $S \rightarrow \text{while } (B)S$

iii) $S \rightarrow \{L\}$

iv) $E \rightarrow \text{id}$

v) $E \rightarrow \text{num}$

vi) $B \rightarrow E > E$

vii) $L \rightarrow SL$

viii) $L \rightarrow \epsilon$

Start Symbol = S

P.T.O.

OR

- Q4)** a) Enlist the operations performed on symbol table. [2]
b) Construct SLR (1) parsing table for the following grammar. [8]

$S \rightarrow aAb|bB$

$A \rightarrow Aa| \epsilon$

$B \rightarrow Bb| \epsilon$

- Q5)** a) What do you mean by common sub-expression? Discuss the algorithm for elimination of common sub-expression. [6]
b) Discuss peephole optimization techniques. [6]
c) What is DAG? with suitable illustrations explain the role of DAG in code generation phase. [6]

OR

- Q6)** a) Discuss following optimizations with example. [6]
i) Strength reduction
ii) Dead code elimination.
b) What do you mean by 'Next Use' information? How it is computed? [6]
c) Explain the algorithm for generating code from labeled tree. [6]
- Q7)** a) Explain the different translation schemes to remove syntactic sugar from Haskell program. [6]
b) Explain following features of object oriented languages related to compiler design. [6]
i) Overloading
ii) Inheritance.
c) Discuss features of Java CC compiler. [4]

OR

- Q8)** a) Discuss following with respect to object oriented languages. [6]
i) Type checking.
ii) Type coercion.
- b) Explain following with respect to functional languages. [6]
i) Referential transparency.
ii) Lazy evaluation.
- c) What is activation record? Explain possible structure of an activation record? [4]
- Q9)** a) Discuss parallel programming models. [6]
b) Write short notes: [6]
i) g++
ii) NVCC
iii) LLVM
- c) Compare processes and threads. [4]

OR

- Q10)** a) Discuss issues in message passing techniques. [6]
b) Explain following concepts related to automatic parallelization. [6]
i) Data dependencies.
ii) Loop transformations.
- c) What is Interpreter? Explain Dalvik. [4]

